

Complete Reaction Mechanism Flowchart for Tamiflu Synthesis

computational methods

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Abstract

Oseltamivir is a shikimate-derived influenza neuraminidase inhibitor whose process development evolved from an early discovery synthesis into industrially optimized Roche routes centered on a stereodefined epoxide intermediate.[1, 2, 5, 6] This paper reformats the supplied notes into a compact technical manuscript organized around an 11-stage flowchart. The focus is mechanistic: esterification and temporary diol protection organize the scaffold for selective activation; intramolecular epoxide formation converts a protected shikimate derivative into a high-value relay intermediate; and controlled nitrogen installation, protection-state management, and late-stage side-chain elaboration converge on oseltamivir or its pharmaceutical phosphate salt. Published process generations are not perfectly identical in step count, telescoping strategy, or azide handling, so the scheme below should be read as a pedagogical map of the route logic rather than a literal transcription of any single manufacturing batch record.[4, 3, 5]

Keywords: oseltamivir; shikimic acid; process chemistry; epoxide intermediate; stereocontrol; influenza neuraminidase inhibitor

carboxylate and became one of the best-known antivirals developed for the treatment of influenza. From a synthetic perspective, its manufacture is notable because the target contains a densely functionalized cyclohexene core with strict stereochemical requirements and a sensitive pattern of oxygen- and nitrogen-bearing substituents. Early discovery chemistry established a practical total synthesis from shikimic acid, whereas subsequent Roche process work emphasized scalable preparation of a shikimic-acid-derived epoxide precursor and more efficient endgame transformations.[1, 2, 5]

The literature now contains several closely related shikimic-acid-based route generations. These include azide-reduced or azide-free variants as well as shorter sequences that minimize protecting-group changes.[3, 4, 6] Accordingly, quoted overall yields and exact step counts vary across sources. In this document, the user-supplied eleven-stage numbering is preserved so that the route can be discussed consistently, but the interpretation is kept broad enough to remain faithful to the Roche-era process literature as a whole.

1 Introduction

Oseltamivir phosphate, marketed as Tamiflu, is the orally delivered prodrug of oseltamivir

2 Reaction Sequence Overview

Figure 1 condenses the supplied sequence into a publication-style flowchart. The labels A–J are retained as convenient handles for discussion, even though published process documents sometimes telescope or rename individual intermediates. Table 1 translates the same map into a mechanistic step-by-step interpretation.

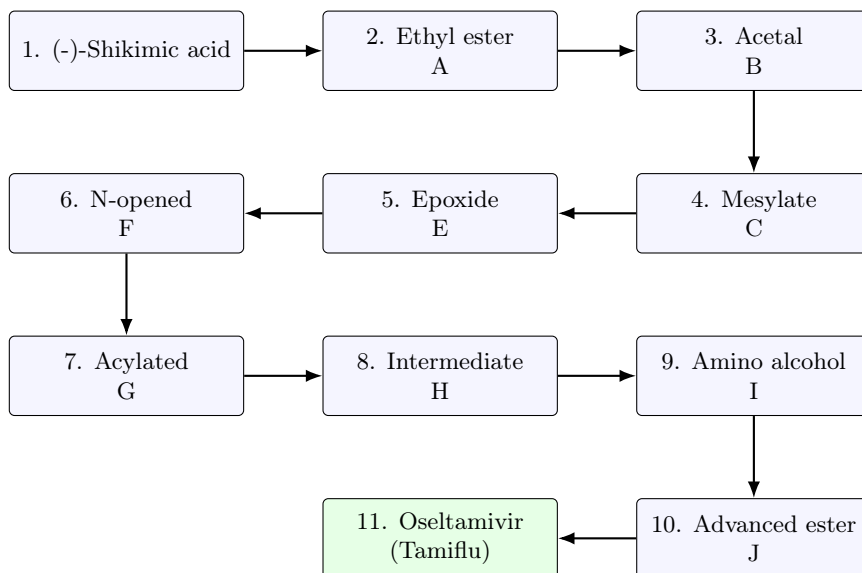


Figure 1: Compact 11-stage flowchart adapted from the supplied oseltamivir pathway. The labels A–J follow the user-provided sequence; Table 1 expands the corresponding transformation logic.

Table 1: Mechanistic interpretation of the supplied 11-stage pathway.

Stage	Transformation	Strategic purpose	Mechanistic comment
1 $\rightarrow$ 2	Carboxyl group to ethyl ester	Converts the acid into a more tractable intermediate for downstream handling	Typically accomplished by acid-promoted alcoholysis or Fischer esterification.
2 $\rightarrow$ 3	3,4-diol protection as pentyldiene acetal	Masks a reactive vicinal diol and biases the conformation of the cyclohexene core	Ketalization suppresses uncontrolled polyfunctional reactivity.
3 $\rightarrow$ 4	Sulfonyl activation of oxygen	Installs a competent leaving group	Mesylation transforms a poor leaving group into one suited for intramolecular displacement.
4 $\rightarrow$ 5	Intramolecular epoxide formation	Generates the strategic relay intermediate used in multiple process generations	Base-promoted anti displacement creates a strained, stereodefined epoxide.
5 $\rightarrow$ 6	Nucleophilic epoxide opening	Introduces the first nitrogen-bearing substituent with stereochemical control	Regioselection is governed by substrate conformation, sterics, and electronics.
6 $\rightarrow$ 7	Nitrogen protection or acylation	Moderates amine reactivity for later manipulations	Acylation improves chemoselectivity and simplifies process control.
7 $\rightarrow$ 8	Late-stage oxygen-side-chain construction	Builds the hydrophobic substituent needed for target activity	The exact implementation varies across route generations and patents.
8 $\rightarrow$ 9	Deprotection and reduction	Reveals an advanced amino alcohol framework	Hydrogenolysis or analogous deprotection chemistry is often used at this stage.
9 $\rightarrow$ 10	Functional-group adjustment	Tunes the oxidation and protection state for the endgame	This stage collects route-dependent cleanup and convergence operations.
10 $\rightarrow$ 11	Final amine unveiling and isolation	Delivers oseltamivir or the corresponding phosphate salt	Endgame steps prioritize purity, stereochemical integrity, and robust isolation.

3 Stepwise Scheme and Mechanistic Discussion

The material below merges the scheme gallery with the mechanistic discussion so that each major step is shown together with its corresponding process rationale. Duplicate uploads were consolidated so that each phase appears only once, and the explanatory note is placed directly beneath the associated image.

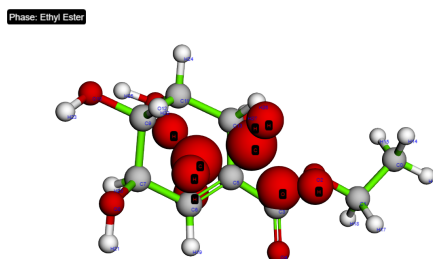


Figure 2. Ethyl ester. Representative rendering of the Fischer-esterified shikimate derivative. Converting the carboxylic acid into the ethyl ester improves handling and prepares the substrate for selective protection and activation in the following stages.

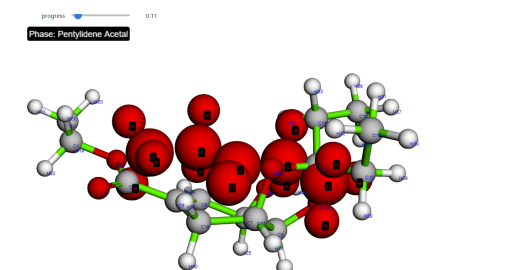


Figure 3. Pentylidene acetal. Representative rendering of the protected intermediate after cyclic ketal formation at the vicinal diol. This masking step suppresses uncontrolled polyhydroxyl reactivity and rigidifies the scaffold for the later intramolecular substitution chemistry.

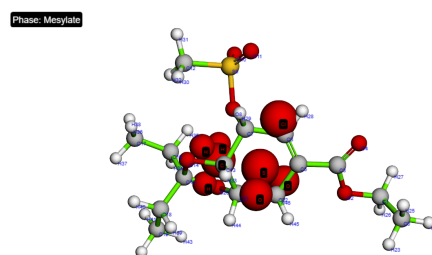


Figure 4. Mesylate. Representative rendering of the sulfonyl-activated intermediate. Mesylation converts a poor leaving group into one suited for the intramolecular displacement that generates the epoxide relay intermediate.

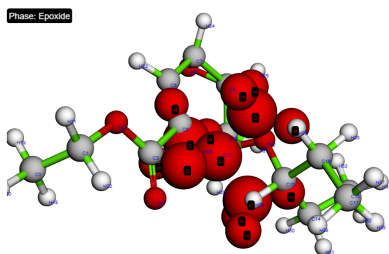


Figure 5. Epoxide. Representative rendering of the strained epoxide intermediate. Base-promoted intramolecular S_N2 closure generates the strategic relay motif that carries stereochemical information into the nitrogen-installation stage.

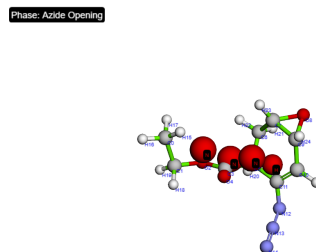


Figure 6. Azide opening. Representative rendering of the nitrogen-installation product formed after regioselective opening of the epoxide. Attack at the less hindered site introduces a nitrogen-bearing substituent while preserving the stereochemical integrity of the cyclohexene core.

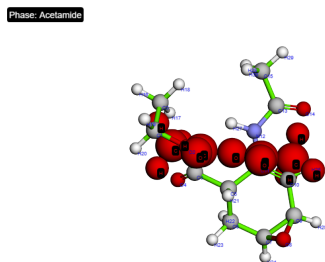


Figure 7. Acetamide. Representative rendering of the acylated amine intermediate. Acylation tempers amine reactivity and improves chemoselective control for the late-stage sequence.

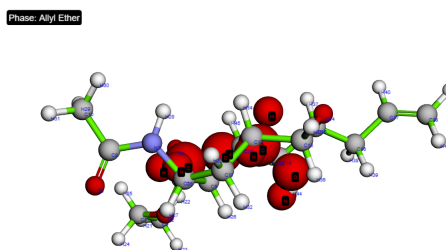


Figure 8. Allyl ether. Representative rendering of the intermediate formed after oxygen-side-chain elaboration. This stage advances pharmacophore construction while preserving the preorganized stereochemical framework.

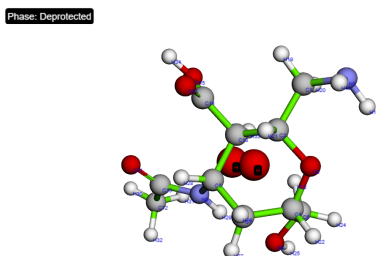


Figure 9. Deprotected intermediate. Representative rendering of the amino-alcohol-like intermediate obtained after removal of temporary protecting groups. Deprotection reveals the advanced functional pattern needed for the final endgame transformations.

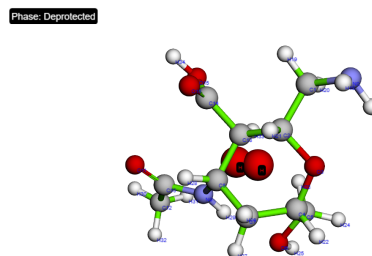


Figure 10. Triol ester. Representative rendering of an advanced ester intermediate with multiple exposed hydroxyl groups. This stage reflects protection-state adjustment immediately before conversion to the drug framework.

Phase: Oseltamivir

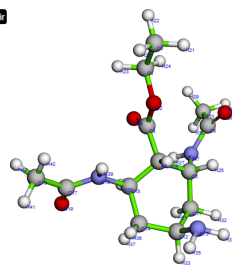


Figure 11. Oseltamivir. Representative rendering of the final target structure. Final unveiling of the amine and side-chain architecture gives the stereochemically defined oseltamivir scaffold.

4 Stereochemical Rationale

The synthetic power of (-)-shikimic acid lies in the fact that it already encodes the relative stereochemistry of the cyclohexene core required for oseltamivir. The route therefore depends more on stereospecific relay operations than on repeated de novo asymmetric induction. Selective protection preserves a favorable conformational preference, intramolecular epoxide formation translates that geometry into a reactive three-membered ring, and nucleophilic opening converts the stored geometric information into the next stereodefined intermediate. This chiron-based strategy is one reason shikimic-acid and quinic-acid approaches dominated early commercial thinking around Tamiflu manufacture.[1, 2, 7]

5 Industrial Significance

From a process viewpoint, the route balances three competing demands: safety, step economy, and stereochemical fidelity. Azides are powerful synthetic handles but create scale-up concerns that motivated alternative nitrogen-installation tactics. Protecting groups enable selectivity but add mass intensity and handling cost, so later generations of the route sought to telescope operations and reduce isolations. At the same time, raw-material supply remained strategically important because shikimic acid can originate from plant extraction, biotransformation of quinic acid, or fermentation-oriented manufacture. These pressures explain why the Roche route became a benchmark not only for antiviral synthesis, but also for broader discussions of process intensification and supply-chain resilience in medicinal chemistry.[3, 4, 6, 7]

6 Computational Representation and Validation Boundaries

The supplied Python sketch based on RDKit is an appropriate starting point for generating a 2D panel of intermediates from validated SMILES strings. It is important, however, to distinguish depiction from mechanistic validation. A rendered molecular grid can demonstrate that a proposed connectivity pattern is chemically parseable and visually consistent, but it does not by itself establish reaction energetics, preferred transition-state approach angles, or agreement with crystallographic reference structures. Those stronger claims require explicit conformer generation, force-field or quantum-chemical calculations, and comparison to experimental structural data. For that reason, the present paper uses a schematic process flowchart and avoids unverified numerical descriptors.

7 Conclusion

This paper converts the supplied code-and-notes bundle into a structured LaTeX manuscript. It describes a shikimic-acid-based route to oseltamivir. Across Roche-era variants, the central process logic is consistent: inherit stereochemical information from the natural-product chiron, manage polarity through temporary protection, create a strategic epoxide relay intermediate, install nitrogen under conformational control, and complete the pharmacophore through late-stage side-chain construction and carefully timed deprotection. The resulting flowchart and table provide a concise reference for medicinal, process, and computational chemists who need a readable overview of the synthesis strategy without reproducing a full proprietary batch protocol.

References

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